

The New India Government — Complete Project Documentation

Independent fact-checking and citizen journalism for a better India.
Built with Next.js 16, Payload CMS 3, PostgreSQL 16, Tailwind CSS 4 and deployed locally via Lando.

Table of Contents

- 1. [Project Overview](#)
- 2. [Tech Stack](#)
- 3. [Local Development — Quick Start](#)
- 4. [All URLs & Credentials](#)
- 5. [Codebase Architecture](#)
- 6. [Collections \(Database Schema\)](#)
- 7. [Pages & Routes](#)
- 8. [API Endpoints](#)
- 9. [Features Deep-Dive](#)
- 10. [Admin Panel Guide](#)
- 11. [Posting News Through AI \(MCP\)](#)
- 12. [Setting Up Claude Code with MCP](#)
- 13. [Environment Variables Reference](#)
- 14. [Production Deployment](#)
- 15. [Security Architecture](#)
- 16. [Troubleshooting & Known Fixes](#)

1. Project Overview

The New India Government is a civic-journalism platform. It is a newspaper-style website where:

- Editors and AI agents **publish fact-checked articles** in six editorial categories
- Citizens **cast live votes** on whether it is time for a new government
- Citizens **sign petitions** targeted at specific government bodies
- Citizens **share their voice** in a moderated public wall
- A scrolling **ticker** shows fact-check credibility scores in real time
- The admin panel doubles as an **AI content studio** via a built-in MCP server that any Claude (or compatible) agent can call to create, update, and manage all content

The design deliberately mimics a broadsheet newspaper — serif display fonts, tight leading, a restrained ink-and-lavender palette, and an editorial layout grid.

2. Tech Stack

Layer	Technology	Version
Frontend framework	Next.js	16.2.x
Rendering	React	19
CMS / backend	Payload CMS	3.84.x
Database	PostgreSQL	16
ORM / adapter	@payloadcms/db-postgres	3.84.x
Rich text editor	Lexical	(via Payload)
CSS framework	Tailwind CSS	4.x

CSS bundler	Turbopack (dev)	via Next.js
Native image processing	sharp	0.33.x
Email catcher (dev)	Mailpit	latest
Local orchestration	Lando	v3.26.x
Container runtime	Docker Desktop	≥ 4.37.x
Package manager	pnpm	9.x
Bot protection	Cloudflare Turnstile	(via @marsidev/react-turnstile)
AI content API	@payloadcms/plugin-mcp	3.84.x
Data fetching (client)	SWR	2.x
Language	TypeScript	5.9.x

3. Local Development — Quick Start

Prerequisites

- **macOS** (arm64 tested; x86 should work)
- **Docker Desktop** ≥ 4.37 installed and running
- **Lando** v3 installed (`~/ .lando/bin/lando`)
- **pnpm** 9 installed globally (`npm i -g pnpm@9`)
- Your user must be in the macOS `admin` group

```
# Check if you are in the admin group
dscacheutil -q group -a name admin | grep users

# If not, add yourself (requires sudo)
sudo dscl . -append /Groups/admin GroupMembership $(whoami)
```

First-time setup

```
# 1 – Clone and enter the project
git clone <repo-url> thenewindiagoverment
cd thenewindiagoverment

# 2 – Update the lockfile to include Linux binaries (needed inside Docker)
# This is already done – just run install to confirm
pnpm install

# 3 – Start the full stack
lando start
# Takes ~2 minutes the first time (Docker images pulled, pnpm install runs inside)

# 4 – Seed the database with demo content
lando seed
```

After the seed completes you will see:

SEED COMPLETE

Admin URL: <https://ngi.lndo.site/admin>
Admin email: admin@thenewindiagov.test
Admin password: ChangeMe!2026
Cowork API key: <generated key>

Daily workflow

```
lando start      # start the stack
lando stop       # pause containers (data is preserved)
lando rebuild    # rebuild containers after config changes
lando destroy    # wipe everything (containers + volumes)
lando seed       # re-seed / top-up demo content (idempotent)
lando pnpm <cmd> # run any pnpm command inside the container
lando psql       # open a psql shell against the payload database
lando tsc        # TypeScript type-check
lando test       # run Vitest unit tests
lando build      # production build (inside container)
```

Useful Lando shortcuts

```
lando logs -s app --follow    # tail the Next.js dev server logs
lando logs -s db --follow     # tail postgres logs
lando logs -s mailpit --follow # tail Mailpit SMTP logs
```

4. All URLs & Credentials

Local development URLs

URL	What it is
https://ngi.lndo.site/	Homepage
https://ngi.lndo.site/admin	Payload CMS admin panel
https://ngi.lndo.site/category/fact-check	Category: Fact-Check
https://ngi.lndo.site/category/policy	Category: Policy
https://ngi.lndo.site/category/elections	Category: Elections
https://ngi.lndo.site/category/leaders	Category: Leaders
https://ngi.lndo.site/category/investigations	Category: Investigations
https://ngi.lndo.site/category/opinion	Category: Opinion
https://ngi.lndo.site/petitions	All active petitions
https://ngi.lndo.site/voices	Wall of citizen voices
<a href="https://ngi.lndo.site/search?q=<term>">https://ngi.lndo.site/search?q=<term>	Full-text search
https://ngi.lndo.site/about	About page
https://ngi.lndo.site/sitemap.xml	Auto-generated sitemap

https://ngi.lndo.site/robots.txt	Robots file
http://localhost:8025	Mailpit email UI (view test emails)
localhost:5433	PostgreSQL (external port for DB tools)

Admin credentials (seeded)

Field	Value
URL	https://ngi.lndo.site/admin
Email	admin@thenewindiagov.test
Password	ChangeMe!2026
Role	Admin (full access)

All seeded users

Email	Password	Role	Capabilities
admin@thenewindiagov.test	ChangeMe!2026	Admin	Everything — create, edit, delete, moderate
editor@thenewindiagov.test	Cowork!2026SecurePassword	Editor	Create & edit articles, petitions
moderator@thenewindiagov.test	Cowork!2026SecurePassword	Moderator	Approve/reject voices, view signatures
cowork@thenewindiagov.test	(API key only)	Cowork	AI agent API access via MCP

Get the Cowork API key: After `lando seed`, check the seed output or run:

```
lando psql
SELECT api_key FROM users WHERE role = 'cowork';
\q
```

Or enable API key in the admin panel: Admin → Users → cowork → enable "API Key".

Database connection (for DB tools like TablePlus, DBeaver)

Field	Value
Host	127.0.0.1
Port	5433
Database	payload
Username	payload
Password	pay load

5. Codebase Architecture

```
thenewindiagovernment/
|
├── .lando.yml           ← Lando dev environment config (Docker orchestration)
```

- └─ next.config.mjs ← Next.js config (CSP, image domains, allowed origins)
- └─ package.json ← Dependencies + pnpm supportedArchitectures
- └─ pnpm-lock.yaml ← Lockfile (includes linux-arm64 binaries for Docker)
- └─ tsconfig.json ← TypeScript config
- └─ config/
 - └─ postgres.conf ← Custom PostgreSQL config
 - └─ postgres-init.sql ← Idempotent role/privilege setup (auto-runs on fresh DB)
- └─ src/
 - └─ payload.config.ts ← MAIN: Payload CMS configuration + MCP plugin
 - └─ collections/ ← Database collections (one file = one table)
 - └─ Articles.ts
 - └─ Authors.ts
 - └─ Categories.ts
 - └─ Tags.ts
 - └─ Media.ts
 - └─ Petitions.ts
 - └─ PetitionSignatures.ts
 - └─ Votes.ts
 - └─ Voices.ts
 - └─ TickerItems.ts
 - └─ AdBanners.ts
 - └─ NewsletterSubscribers.ts
 - └─ Users.ts
 - └─ globals/
 - └─ SiteSettings.ts ← Site-wide settings (vote question, tagline, etc.)
 - └─ access/ ← Reusable Payload access-control functions
 - └─ isAdmin.ts ← True only for role=admin
 - └─ isAdminOrEditor.ts ← True for admin or editor
 - └─ isAdminOrSelf.ts ← True for admin, or the user themselves
 - └─ publicRead.ts ← True always (public read)
 - └─ hooks/
 - └─ revalidateAfterChange.ts ← After DB write → calls /api/revalidate → clears Next.js
- cache
 - └─ lib/ ← Shared utilities
 - └─ format.ts ← Date formatting helpers
 - └─ hash.ts ← SHA-256 (used for voter dedup hashes)
 - └─ ratelimit.ts ← In-memory token-bucket rate limiter
 - └─ seo.ts ← generateBaseMetadata / generateArticleMetadata
 - └─ turnstile.ts ← Cloudflare Turnstile server-side verification
 - └─ seed/
 - └─ seed.ts ← Idempotent demo-data seeder
 - └─ app/
 - └─ globals.css ← CSS custom properties (design tokens) + Tailwind
 - └─ fonts.ts ← Font definitions (display, ui, body)
 - └─ robots.ts ← /robots.txt route
 - └─ sitemap.ts ← /sitemap.xml route (dynamic, from DB)
 - └─ (frontend)/ ← Public-facing website (Next.js route group)

- └─ layout.tsx ← Masthead + Ticker + Nav + Footer wrapper
- └─ page.tsx ← Homepage
- └─ about/page.tsx
- └─ search/page.tsx
- └─ petitions/page.tsx
- └─ voices/page.tsx
- └─ article/[slug]/page.tsx
- └─ category/[slug]/page.tsx
- └─ petition/[slug]/page.tsx
- └─ (payload)/ ← Payload CMS admin panel (Next.js route group)
 - └─ admin/[...segments]/page.tsx
- └─ api/ ← REST API routes
 - └─ [...slug]/route.ts ← Payload's built-in REST & GraphQL API
 - └─ vote/route.ts ← GET live vote tallies / POST cast a vote
 - └─ voice/route.ts ← POST submit a voice
 - └─ newsletter/subscribe/route.ts ← POST subscribe email
 - └─ petition/sign/route.ts ← POST sign a petition
 - └─ revalidate/route.ts ← POST clear Next.js page cache
- └─ components/
 - └─ ads/SideBanner.tsx
 - └─ article/
 - └─ ArticleCard.tsx ← Reusable article card (3 variants: left/right/default)
 - └─ EditorialImage.tsx ← Next.js Image wrapper
 - └─ FeaturedHero.tsx ← Large homepage hero article
 - └─ grid/HomeGrid.tsx ← 3-column editorial grid layout
 - └─ masthead/
 - └─ Badge.tsx ← The ornate chakra badge SVG
 - └─ Masthead.tsx ← Full newspaper masthead header
 - └─ nav/PrimaryNav.tsx ← Category navigation bar
 - └─ newsletter/SignupCard.tsx
 - └─ petition/
 - └─ PetitionCard.tsx ← Petition summary card with progress bar
 - └─ SignForm.tsx ← Petition signing form (client component)
 - └─ ticker/FactCheckTicker.tsx ← Scrolling fact-check score marquee
 - └─ ui/Divider.tsx
 - └─ vote/
 - └─ VoteWidget.tsx ← Live poll (client component, SSE tally via SWR)
 - └─ VoiceModal.tsx ← Post-vote popup: share your opinion

6. Collections (Database Schema)

Each collection in `src/collections/` maps to a PostgreSQL table managed automatically by Payload. Payload auto-applies schema changes in development (`push: true`).

Articles

The heart of the site. Supports drafts, autosave, and versioning.

Field	Type	Notes
title	text (localized)	Max 180 chars
slug	text (unique)	URL slug, kebab-case

kicker	text (localized)	e.g. "FACT-CHECK", "INVESTIGATION"
excerpt	textarea (localized)	Max 320 chars — shown in cards
heroImage	upload → Media	Optional
body	richText (Lexical)	Full article body
category	relationship → Categories	Required
tags	relationship → Tags	Many
author	relationship → Authors	Required
credibilityScore	number (0–10)	Shown in ticker; 10 = fully accurate
sources	array of {label, url}	Linked citations
featured	checkbox	Shows in homepage hero
placement	select	feed / homepage-left / homepage-right / homepage-hero
status	select	draft / published / archived
publishedAt	date	
seo	group	metaTitle, metaDescription, ogImage

Access: Public read of published articles. Admin + Editor can create/edit. Only Admin can delete.

Hook: After every save → calls `POST /api/revalidate?tag=article` → Next.js clears the `article` cache tag.

Categories

The six editorial sections.

Field	Type
name	text
slug	text (unique)
description	textarea
accentColor	text (hex)

Seeded slugs: `fact-check`, `policy`, `elections`, `leaders`, `investigations`, `opinion`

Authors

Journalist/contributor profiles.

Field	Type
name	text
slug	text (unique)
bio	textarea
avatar	upload → Media

twitter	text
email	email

Seeded authors: Priya Sharma, Rahul Menon, Ananya Krishnan, Vikram Joshi

Tags

Free-form keyword tags attached to articles.

Field	Type
name	text
slug	text (unique)

Media

File/image uploads. Processed by `sharp` on upload to generate sized variants: `thumbnail` , `card` , `hero` .

Access: Public read. Admin + Editor can upload.

Petitions

Public civic petitions targeted at government bodies.

Field	Type	Notes
title	text	
slug	text (unique)	
summary	textarea	Shown in cards
body	richText	Full petition text
target	text	e.g. "Prime Minister's Office"
goalSignatures	number	Default 1000
currentSignatures	number	Auto-updated by signature hook
heroImage	upload → Media	
status	select	active / closed / draft

PetitionSignatures

Each signature on a petition.

Field	Type	Notes
petition	relationship → Petitions	
firstName	text	Max 60 chars
lastName	text	Max 60 chars
city	text	

country	text	Default "India"
voterHash	text (unique)	SHA-256(salt + petitionId + IP + UA) for dedup
comment	textarea	Optional public comment
displayPublic	checkbox	If true, shown on petition page
verified	checkbox	Default true

Hook: After each new signature → recounts and updates `petition.currentSignatures` .

Access: Admin + Moderator only (signatures are private). Public `create` for the sign API.

Votes

Each cast vote on the main poll.

Field	Type	Notes
option	text	e.g. yes-time-for-change
voterHash	text (unique)	SHA-256(salt + IP + UA) — one vote per device
userAgent	text	
ipCountry	text	From Cloudflare header

Access: Admin read only. Public create (via API).

Voices

Citizen statements on the Wall of Voices.

Field	Type	Notes
text	textarea	Max 1000 chars
displayName	text	Optional, defaults to "Anonymous"
status	select	pending / approved / rejected
voteOption	select	Optional — links voice to poll position

Access: Admin + Moderator can read/moderate. Public create (via API). Only approved voices appear on the wall.

TickerItems

Items shown in the scrolling fact-check ticker at the top of every page.

Field	Type	Notes
claim	text	Short government claim to evaluate
verdict	select	misleading / false / mostly-false / mixed / true
credibilityScore	number (0–10)	
linkedArticle	relationship → Articles	Click through to full article

active	checkbox	Show/hide without deleting
order	number	Display order (ascending)

AdBanners

Promotional banners (sponsorships, campaigns).

Field	Type	Options
label	text	
image	upload → Media	
link	text	URL
placement	select	homepage-left / homepage-right / article-inline
active	checkbox	

NewsletterSubscribers

Email newsletter sign-ups.

Field	Type
email	email (unique)
source	text (default: "homepage")

Access: Admin read only. Public create (via API). Deduplication by email.

Users

Internal staff accounts.

Field	Type	Options
email	email	
password	(hashed)	
role	select	admin / editor / moderator / cowork
apiKey	text	Auto-generated by Payload when enabled

Roles:

- **Admin** — Full access to everything
- **Editor** — Create and edit articles, petitions
- **Moderator** — Approve/reject voices, view petition signatures
- **Cowork** — API-key only access for AI agents via MCP

SiteSettings (Global)

Site-wide configuration stored as a single record in the DB.

Field	Default value	What it controls
-------	---------------	------------------

siteName	"The New India Government"	
tagline	"Truth · Transparency · Voice"	Shown in masthead + footer
badgeText	"THE CITIZEN'S PRESS"	
cityLine	"NEW DELHI · MUMBAI · BENGALURU"	
voteQuestion	"Is it time for a new India government?"	The poll question
voteOptions	YES / NO / UNDECIDED	The 3 poll buttons
footer.aboutText	—	Footer description
footer.links	—	Footer link array

Edit at: **Admin → Site Settings**

7. Pages & Routes

Frontend pages

Route	Component	Description
/	page.tsx	Homepage: 3-column grid, poll, active petitions
/about	about/page.tsx	About the publication
/search?q=<term>	search/page.tsx	Full-text article search
/petitions	petitions/page.tsx	All active petitions list
/voices	voices/page.tsx	Wall of approved citizen voices
/category/[slug]	category/page.tsx	Articles filtered by category
/article/[slug]	article/page.tsx	Full article, related stories, credibility score
/petition/[slug]	petition/page.tsx	Petition detail + sign form + signers
/admin/*	Payload panel	CMS admin interface
/sitemap.xml	sitemap.ts	Auto-generated XML sitemap
/robots.txt	robots.ts	Disallows /admin and /api

Layout components on every page

Every frontend page is wrapped in `(frontend)/layout.tsx` which renders:

1. **Masthead** — Newspaper-style header with title, date, ornament
 2. **FactCheckTicker** — Scrolling marquee of fact-check scores
 3. **PrimaryNav** — Category navigation bar
 4. **{children}** — Page content
 5. **Footer** — Links, copyright
-

8. API Endpoints

Built-in Payload REST API

Payload automatically exposes a full REST API at `/api/<collection>` :

```
GET    /api/articles      ← List articles (public)
GET    /api/articles/:id  ← Get one article
POST   /api/articles      ← Create (requires auth)
PATCH /api/articles/:id ← Update (requires auth)
DELETE /api/articles/:id ← Delete (admin only)
POST   /api/users/login ← Login → JWT cookie
POST   /api/users/logout
GET    /api/globals/site-settings
```

Custom public-facing API routes

GET `/api/vote`

Returns live poll tallies.

```
{
  "totals": {
    "yes-time-for-change": 142,
    "no-current-fine": 51,
    "undecided": 23
  },
  "total": 216
}
```

POST `/api/vote`

Cast a vote.

```
// Request
{ "option": "yes-time-for-change", "turnstileToken": "<token>" }

// Response (success)
{ "ok": true }

// Response (already voted – same device)
{ "ok": true, "already": true } // HTTP 409
```

Rate-limited: 10 requests/minute per IP. Deduplicated by `SHA256(salt + IP + UA)` .

POST `/api/voice`

Submit a voice to the Wall of Voices (goes to `pending` , needs moderator approval).

```
// Request
{
  "text": "The government must be honest about economic data.",
  "displayName": "Ravi Kumar", // optional
  "option": "yes-time-for-change", // optional, links to poll vote
  "turnstileToken": "<token>"
}

// Response
{ "ok": true }
```

POST /api/newsletter/subscribe

Subscribe to the newsletter.

```
// Request
{ "email": "citizen@example.com" }

// Response (success or already subscribed)
{ "ok": true }
{ "ok": true, "already": true }
```

POST /api/petition/sign

Sign a petition.

```
// Request
{
  "petitionId": 1,           // numeric ID or string "1"
  "firstName": "Priya",
  "lastName": "Sharma",
  "city": "Mumbai",         // optional
  "comment": "Support this!", // optional
  "displayPublic": true,    // optional, default true
  "turnstileToken": "<token>"
}

// Response
{ "ok": true }
{ "ok": true, "already": true } // already signed from same device
```

POST /api/revalidate?tag=<tag>&secret=<secret>

Clears the Next.js cache for a specific tag. Called automatically by Payload hooks when content is saved.

```
curl -X POST "https://ngi.lndo.site/api/revalidate?tag=article&secret=c4e7f0a9..."
```

POST /api/mcp (built by @payloadcms/plugin-mcp)

The AI content API. See Section 11 for full details.

9. Features Deep-Dive

Fact-Check Ticker

A continuous horizontal marquee of claims and their credibility scores (0–10). Rendered server-side on every page load. Clicking a claim navigates to the linked article.

To add a ticker item: Admin → Ticker Items → Add New

Or via AI: see Section 11.

The People's Poll

A full-width client-side poll widget on the homepage (#vote). Features:

- **Live vote counts** — updates every 5 seconds via SWR polling of GET /api/vote
- **Cloudflare Turnstile** protection (invisible challenge, auto-passes in dev)

- **Deduplication** — one vote per device (hashed IP + User-Agent, not stored as plain text)
- **Post-vote modal** — after voting, a dialog invites the user to share their voice (which goes to the Wall of Voices pending moderation)
- **Animated progress bars** — percentage bars animate into view after vote is cast

Customise the poll question and options: Admin → Site Settings

Wall of Voices

A masonry two-column grid of approved citizen statements. Voices are colour-coded by poll position (lavender = YES, dark = NO, grey = UNDECIDED).

Workflow:

1. Citizen votes → post-vote modal → submits voice
 2. Voice saved with `status: pending`
 3. Moderator logs in → Admin → Voices → approve or reject
 4. Approved voices appear on `/voices`
-

Petitions

Each petition has:

- A title, summary, and rich-text body
 - A progress bar (`currentSignatures` / `goalSignatures`)
 - A "Sign This Petition" sidebar form with Turnstile protection
 - A list of recent public signers (name, city, optional comment)
 - Signature counts auto-update in the DB after each new signature
-

Full-text Search

`/search?q=<term>` queries the `articles` table using Payload's built-in search, filtering to `status: published` only. Results are returned server-side and rendered as article cards.

SEO & Structured Data

Every page has:

- `<title>` and `<meta name="description">` set from DB content
 - Open Graph and Twitter Card tags
 - `NewsArticle` JSON-LD schema on article pages
 - `WebSite` + `Organization` JSON-LD on the homepage
 - Dynamic `sitemap.xml` covering all articles, petitions, categories
 - `robots.txt` allowing all except `/admin` and `/api`
-

Incremental Static Regeneration (ISR)

Pages use `export const revalidate = 60` — Next.js serves a cached version and regenerates in the background every 60 seconds. Additionally, any save in the Payload admin immediately calls `POST /api/revalidate` to clear the relevant tag so the next visitor always gets the latest content.

Bilingual Support (English / Hindi)

Articles, titles, kickers, and excerpts are marked `localized: true` in the schema. Payload handles locale routing. Hindi translations can be added via the admin panel on any article field.

10. Admin Panel Guide

Accessing the panel

Go to `https://ngi.lndo.site/admin` and sign in with your credentials.

Publishing an article

1. **Admin → Articles → Create New**
2. Fill in:
 - **Title** — The headline (max 180 chars)
 - **Slug** — URL-friendly identifier (kebab-case , e.g. `gdp-data-manipulation`)
 - **Kicker** — Short label like `FACT-CHECK` or `INVESTIGATION` (shown in red above headline)
 - **Excerpt** — 1–2 sentence teaser (max 320 chars)
 - **Category** — Select one of the six categories
 - **Author** — Select an author profile
 - **Body** — Full article in the Lexical rich-text editor (headings, lists, blockquotes, links)
 - **Credibility Score** — 0 (misleading) to 10 (accurate) — appears in the ticker
 - **Sources** — Add cited links
 - **Placement** — Set to `homepage-hero` for the lead story, `homepage-left` for the secondary column
 - **Status** — Change from `draft` to `published`
 - **Published At** — Set the date
3. Click **Save** — the page cache is automatically cleared

Moderating voices

1. **Admin → Voices**
2. Filter by `status: pending`
3. Open each voice → change status to `approved` (shows on wall) or `rejected` (hidden)

Managing the ticker

1. **Admin → Ticker Items → Create New**
2. Enter the government claim, select a verdict, enter a credibility score (0–10), optionally link an article
3. Set `active: true` and assign an `order` number
4. Save — appears in the scrolling marquee immediately on next page load

Configuring the site

1. **Admin → Site Settings**
2. Change the vote question, vote options, tagline, footer text

11. Posting News Through AI (MCP)

The platform ships with `@payloadcms/plugin-mcp` which exposes an **MCP (Model Context Protocol) server** at:

`https://ngi.lndo.site/api/mcp`

Any AI agent (Claude, Cursor, etc.) can connect to this endpoint and use natural language to create, update, and manage all site content without touching the admin panel.

What the AI can do via MCP

Collection	Find	Create	Update	Delete
------------	------	--------	--------	--------

articles	✓	✓	✓	✗
media	✓	✓	✗	✗
categories	✓	✗	✗	✗
authors	✓	✗	✗	✗
tags	✓	✓	✗	✗
ticker-items	✓	✓	✓	✓
voices	✓	✗	✓	✗
petitions	✓	✓	✓	✗

Getting the API key

The API key belongs to the `cowork` user (role: Cowork). To get/generate it:

Option A — via Admin panel (recommended)

1. Go to `https://ngi.lndo.site/admin`
2. Sign in as Admin
3. Navigate to **Admin** → **Users** → cowork@thenewindiagov.test
4. Scroll to **API Key** section → click **Generate Key**
5. Copy the key — it is shown only once

Option B — via psql

```
lando psql
SELECT api_key FROM users WHERE role = 'cowork';
\q
```

Option C — from seed output (run after `lando seed`) The key is printed in the terminal and written to `.cowork.key`.

Using MCP from a REST client

You can call MCP directly with `curl` to test it:

```
# List all published articles
curl -X POST https://ngi.lndo.site/api/mcp \
  -H "Authorization: Bearer <YOUR_COWORK_API_KEY>" \
  -H "Content-Type: application/json" \
  -d '{
    "jsonrpc": "2.0",
    "id": 1,
    "method": "tools/call",
    "params": {
      "name": "find-articles",
      "arguments": { "where": { "status": { "equals": "published" } }, "limit": 5 }
    }
  }'
```

```
# Create a new article
curl -X POST https://ngi.lndo.site/api/mcp \
  -H "Authorization: Bearer <YOUR_COWORK_API_KEY>" \
```



```
-H "Content-Type: application/json" \
-d '{
  "jsonrpc": "2.0",
  "id": 2,
  "method": "tools/call",
  "params": {
    "name": "create-articles",
    "arguments": {
      "data": {
        "title": "India's Education Budget Fraud Exposed",
        "slug": "education-budget-fraud-exposed",
        "kicker": "INVESTIGATION",
        "excerpt": "Internal ministry documents show the education budget has been
systematically underfunded for five years.",
        "category": 5,
        "author": 1,
        "status": "published",
        "publishedAt": "2026-05-13T12:00:00.000Z",
        "placement": "feed",
        "credibilityScore": 8.5
      }
    }
  }
}
```

12. Setting Up Claude Code with MCP

Claude Code can connect to the MCP server and post articles, update the ticker, manage petitions, and more — all through natural language.

Step 1 — Install Claude Code

```
npm install -g @anthropic-ai/claude-code
# or
brew install claude-code
```

Authenticate:

```
claude login
```

Step 2 — Add the MCP server to Claude Code

```
# Run this command (replace <KEY> with your cowork API key)
claude mcp add new-india-gov \
  --url https://ngi.lndo.site/api/mcp \
  --header "Authorization: Bearer <YOUR_COWORK_API_KEY>"
```

Or add it manually to `~/.claude/settings.json` :

```
{
  "mcpServers": {
    "new-india-gov": {
```

```
    "url": "https://ngi.lndo.site/api/mcp",
    "headers": {
      "Authorization": "Bearer <YOUR_COWORK_API_KEY>"
    }
  }
}
```

Step 3 — Use it

Open Claude Code in any directory and start a conversation:

```
claude

> "Publish a new fact-check article about the government's inflation data.
> Title: 'RBI Inflation Numbers Don't Match Ground Reality'.
> Category: fact-check. Author: Rahul Menon.
> Credibility score 3.5. Add it to the homepage left column."
```

Claude will:

1. Call `find-categories` to get the `fact-check` category ID
2. Call `find-authors` to get Rahul Menon's ID
3. Call `create-articles` with all the data
4. Report the published article URL

Step 4 — Example AI workflows

Daily news briefing:

```
"Find all articles published in the last 7 days and summarise them in a briefing."
```

Batch ticker update:

```
"Add 5 new ticker items for the government's latest claims about GDP,
infrastructure spending, and employment. Score each based on available facts."
```

Petition creation:

```
"Create a new petition titled 'Demand Full RTI Compliance from SEBI'.
Target: Securities and Exchange Board of India. Goal: 5000 signatures."
```

Voice moderation assistance:

```
"Find all pending voices and help me decide which ones to approve –
show me the text and suggest yes/no for each."
```

Using with Cursor / Windsurf / other MCP clients

Add to your editor's MCP config:

```
{
  "mcpServers": {
    "new-india-gov": {
      "url": "https://ngi.lndo.site/api/mcp",
      "headers": {
```

```

    "Authorization": "Bearer <YOUR_COWORK_API_KEY>"
  }
}
}
}

```

13. Environment Variables Reference

All environment variables are set in `.lando.yml` for local development. For production, set these in your hosting provider's dashboard.

Required variables

Variable	Local dev value	Description
DATABASE_URI	postgres://payload:payload@db/payload	PostgreSQL connection string
PAYLOAD_SECRET	b3f8a2d1...	64-char secret for JWT signing. CHANGE IN PRODUCTION
PAYLOAD_PUBLIC_SERVER_URL	https://ngi.lndo.site	Full URL of the site (no trailing slash)
NEXT_PUBLIC_SITE_URL	https://ngi.lndo.site	Same — used in client-side code

Cloudflare Turnstile (bot protection)

Variable	Local dev value	Description
NEXT_PUBLIC_TURNSTILE_SITE_KEY	1x00000000000000000000AA	Client-side widget key. Dev value always passes
TURNSTILE_SECRET_KEY	1x000000000000000000000000000000AA	Server-side verify key. Dev value always passes

Production: Get real keys from [Cloudflare Turnstile Dashboard](#)

Security tokens

Variable	Local dev value	Description
VOTE_SALT	b3f8a2d1c4e7f0a9b2c5d8e1f4a7b0c3	Salt for voter-dedup hashes. Change before launch
REVALIDATE_SECRET	c4e7f0a9b2c5d8e1f4a7b0c3d6e9f2a5	Secret for the cache-revalidation webhook

Email / SMTP

Variable	Local dev value	Description
SMTP_HOST	mailpit	Dev: Mailpit. Prod: your SMTP provider
SMTP_PORT	1025	Dev: 1025. Prod: 587 (TLS) or 465 (SSL)
SMTP_FROM	The New India Government <noreply@ngi.lndo.site>	From address

Seed credentials

Variable	Value	Description
----------	-------	-------------

SEED_ADMIN_EMAIL	admin@thenewindia.gov.test	Admin user created by seed
SEED_ADMIN_PASSWORD	ChangeMe!2026	Admin password. Change in production
SEED_COWORK_EMAIL	cowork@thenewindia.gov.test	AI agent user

14. Production Deployment

Supported hosting options

The app outputs a **Next.js standalone build** (`output: 'standalone'`), which means it's portable to any Node.js server.

Platform	Difficulty	Notes
Railway	★ Easiest	Native Docker + Postgres managed DB
Render	★ Easiest	Web service + managed Postgres
Fly.io	★★ Medium	Docker-based, global edge
DigitalOcean App Platform	★★ Medium	Managed Postgres included
VPS (any)	★★★ Manual	Full control, Docker Compose
Vercel	★★ Medium	Serverless — needs external Postgres

Option A — Railway (recommended fastest)

1. Create a [Railway](#) account and new project
2. Add a **PostgreSQL** database service — copy the `DATABASE_URL`
3. Create a **Web Service** from your GitHub repo
4. Set the build command: `pnpm install && pnpm build`
5. Set the start command: `node .next/standalone/server.js`
6. Add all environment variables (see Section 13)
7. Set `DATABASE_URI` to the Railway Postgres URL
8. Set `PAYLOAD_PUBLIC_SERVER_URL` and `NEXT_PUBLIC_SITE_URL` to your Railway URL
9. Deploy

Option B — VPS with Docker Compose

Create a `docker-compose.prod.yml` :

```
version: '3.9'
services:
  db:
    image: bitnami/postgresql:16
    environment:
      POSTGRESQL_DATABASE: payload
      POSTGRESQL_USERNAME: payload
      POSTGRESQL_PASSWORD: ${DB_PASSWORD}
    volumes:
      - db_data:/bitnami/postgresql
      - ./config/postgres-init.sql:/docker-entrypoint-initdb.d/init.sql
    restart: unless-stopped

  app:
```

```

build: .
environment:
  DATABASE_URI: postgres://payload:${DB_PASSWORD}@db/payload
  PAYLOAD_SECRET: ${PAYLOAD_SECRET}
  PAYLOAD_PUBLIC_SERVER_URL: https://yourdomain.com
  NEXT_PUBLIC_SITE_URL: https://yourdomain.com
  NEXT_PUBLIC_TURNSTILE_SITE_KEY: ${TURNSTILE_SITE_KEY}
  TURNSTILE_SECRET_KEY: ${TURNSTILE_SECRET}
  VOTE_SALT: ${VOTE_SALT}
  REVALIDATE_SECRET: ${REVALIDATE_SECRET}
  SMTP_HOST: ${SMTP_HOST}
  SMTP_PORT: ${SMTP_PORT}
  SMTP_FROM: ${SMTP_FROM}
  NODE_ENV: production
ports:
  - "3000:3000"
depends_on:
  - db
restart: unless-stopped

volumes:
  db_data:

```

```

# Dockerfile (add to project root)
FROM node:22-alpine AS base
RUN npm install -g pnpm@9

FROM base AS deps
WORKDIR /app
COPY package.json pnpm-lock.yaml ./
RUN pnpm install --frozen-lockfile

FROM base AS builder
WORKDIR /app
COPY --from=deps /app/node_modules ./node_modules
COPY . .
RUN pnpm build

FROM base AS runner
WORKDIR /app
ENV NODE_ENV=production
COPY --from=builder /app/.next/standalone ./
COPY --from=builder /app/.next/static ./next/static
COPY --from=builder /app/public ./public
EXPOSE 3000
CMD ["node", "server.js"]

```

Deploy:

```

# On your VPS
git clone <repo>
cd thenewindia government
cp .env.example .env.production # fill in all vars
docker compose -f docker-compose.prod.yml up -d

```

```
# Run migrations and seed
docker compose exec app node -e "
  const { getPayload } = require('payload');
  // ... or use pnpm seed inside the container
"
```

Option C — Render

1. Create a **PostgreSQL** database on Render → copy the Internal Database URL
2. Create a **Web Service** — connect your GitHub repo
3. Build command: `pnpm install && pnpm build`
4. Start command: `node .next/standalone/server.js`
5. Add environment variables
6. Deploy

Post-deployment checklist

- ☐ Change `PAYLOAD_SECRET` to a new 64-character random string
- ☐ Change all seed passwords
- ☐ Get real Cloudflare Turnstile keys (replace dev test keys)
- ☐ Generate new `VOTE_SALT` and `REVALIDATE_SECRET`
- ☐ Set up a real SMTP provider (Resend, SendGrid, Postmark)
- ☐ Configure your domain/DNS
- ☐ Set up HTTPS (automatic on Railway/Render, or use Caddy/nginx on VPS)
- ☐ Run `pnpm seed` (or equivalent) in production to create admin user
- ☐ Log in and change admin password immediately
- ☐ Generate the Cowork API key for AI agents

Making the MCP available to AI over the internet

Once deployed, your MCP server is at:

```
https://yourdomain.com/api/mcp
```

Update the Claude Code / Cursor config with the production URL and a new API key. No other changes needed — the MCP plugin is built into the app.

15. Security Architecture

Cloudflare Turnstile

All user-facing forms (vote, voice, petition sign, newsletter) require a valid Turnstile token. The token is verified server-side before any data is written to the database.

Voter deduplication

Votes and petition signatures are deduplicated using:

```
SHA-256(VOTE_SALT + petitionId + clientIP + userAgent)
```

The hash is stored; the original IP is never persisted.

Rate limiting

All POST API endpoints are protected by an in-memory token-bucket rate limiter: **10 requests per minute per IP hash**. Implemented in `src/lib/ratelimit.ts`.

Content Security Policy

Strict CSP headers on every response:

- `script-src` — self + Cloudflare Challenges only
- `frame-src` — Cloudflare Challenges only
- `img-src` — self + data: + blob: (no remote image domains in production)
- `connect-src` — self + Cloudflare Challenges only

Access control layers

- **Public routes** — Articles, Categories, Authors, Petitions, Tags: read-only
- **Protected routes** — Voices, Votes, Signatures, Newsletter Subscribers: admin/moderator read
- **Write routes** — Articles/Petitions: Admin + Editor only
- **Admin-only** — Delete any collection, manage users

Secrets

- `PAYLOAD_SECRET` — used by Payload to sign JWT session cookies (must be 64+ chars in production)
- `REVALIDATE_SECRET` — prevents unauthorised cache flushes
- `VOTE_SALT` — prevents rainbow-table attacks on voter hashes

robots.txt

Disallows `/admin/` and `/api/` from search engine crawlers.

16. Troubleshooting & Known Fixes

Lando won't start — port 5432 already in use

The `portforward: 5433` setting maps to external port 5433. If that fails too:

```
# Check what is using the port
lsof -i :5433
# Kill it, or change portforward value in .lando.yml
```

"User does not have permission to install the build engine"

Your macOS user is not in the `admin` group:

```
sudo dscl . -append /Groups/admin GroupMembership $(whoami)
```

Postgres role "payload" does not exist (fresh `lando destroy` + `lando start`)

This is handled automatically by `config/postgres-init.sql` which is mounted to `/docker-entrypoint-initdb.d/init.sql`. It runs on first DB initialisation and creates the `payload` role idempotently.

If you still hit this error, run manually:

```
lando psql # connects as postgres superuser
# then:
CREATE ROLE payload WITH LOGIN PASSWORD 'payload';
GRANT ALL PRIVILEGES ON DATABASE payload TO payload;
```

```
ALTER DATABASE payload OWNER TO payload;
\q
```

Vote buttons are unclickable

Two possible causes:

1. **Already voted** — clear `localStorage` in your browser (DevTools → Application → Local Storage → delete `ngi:voted`)
2. **Turnstile not loading** — in dev, the widget auto-bypasses. In production, ensure Cloudflare's CDN is reachable and CSP headers allow `challenges.cloudflare.com`

"Cannot find module lightningcss.linux-arm64-gnu.node"

The `pnpm-lock.yaml` was generated on macOS and is missing Linux binaries. Fix:

```
# On your macOS host (not inside container)
pnpm install # updates lockfile with linux-arm64 binaries
lando rebuild # reinstalls inside container with new lockfile
```

Mailpit not running after `lando start`

Confirmed fixed — `command: /mailpit` was added to the mailpit service in `.lando.yml` . If it recurs:

```
lando logs -s mailpit
```

Next.js HMR not working from `ngi.lndo.site`

Confirmed fixed — `allowedDevOrigins: ['ngi.lndo.site']` is in `next.config.mjs` .

Voices page shows "No voices yet" despite approved voices in DB

Confirmed fixed — `overrideAccess: true` was added to the `payload.find` call in `voices/page.tsx` . The `Voices` collection `read` access is admin-only, so server components need to explicitly override access control.

Petition signatures not showing on petition page

Same fix — `overrideAccess: true` was added to the `petition-signatures` query in `petition/[slug]/page.tsx` .

Duplicate vote/petition-sign returns 500 instead of 409

Confirmed fixed — Payload v3 wraps PostgreSQL's `23505` unique-violation error in a `ValidationError` . The catch blocks in `vote/route.ts` and `petition/sign/route.ts` now check for `err.name === 'ValidationError'` && `/voterHash/.test(String(e))` in addition to the raw `23505` code.

Quick-Reference Card

THE NEW INDIA GOVERNMENT — Dev Quick Reference

SITE	<code>https://ngi.lndo.site/</code>
ADMIN	<code>https://ngi.lndo.site/admin</code>
EMAIL UI	<code>http://localhost:8025</code>
MCP SERVER	<code>https://ngi.lndo.site/api/mcp</code>

ADMIN LOGIN

Email: admin@thenewindiagov.test
Password: ChangeMe!2026

DATABASE 127.0.0.1:5433 / db=payload / user=payload / pw=payload

COMMANDS

lando start	Start everything
lando stop	Stop containers
lando rebuild -y	Rebuild after .lando.yml changes
lando seed	Seed / refresh demo data
lando pnpm <cmd>	Run pnpm inside container
lando psql	Open postgres shell
lando tsc	TypeScript check
lando test	Run unit tests

Last updated: 2026-05-13 — generated from the running Lando environment.